Integrale curbilinii

Să se calculeze următoarele integrale curbilinii de prima speță:

① $\int_\gamma (x^2+y^2)\ln z\,ds$, unde parametrizarea curbei este

$$\gamma(t) = \begin{cases} x(t) = e^t\cos t \\ y(t) = e^t \sin t \\ z(t) = e^t \end{cases} ; t\in[0,1]$$

$$\gamma'(t) = \begin{cases} x'(t) = e^t\cos t - e^t\sin t = e^t(\cos t - \sin t) \\ y'(t) = e^t\sin t + e^t\cos t = e^t(\sin t + \cos t) \\ z'(t) = e^t \end{cases} ; t\in[0,1]$$

$$\|\gamma'(t)\| = \sqrt{e^{2t}(\cos^2 t - 2\cos t\sin t + \sin^2 t) + e^{2t}(\sin^2 t + \cos^2 t + \sin 2t) + e^{2t}}$$

$$= \sqrt{e^{2t} + \cancel{e^{2t}\sin 2t} - \cancel{e^{2t}\sin 2t} + e^{2t} + e^{2t}} = \sqrt{3e^{2t}} = e^t\sqrt{3}.$$

$$\Rightarrow \int_\gamma (x^2+y^2)\ln z\,ds = \int_0^1 e^t\sqrt{3}\,(e^{2t}\sin^2 t + e^{2t}\cos^2 t)\cdot t\,dt =$$

$$= \int_0^1 t\cdot e^{3t}\cdot\sqrt{3}\,dt = \sqrt{3}\int_0^1 \frac{1}{3}t\cdot 3e^{3t}dt = \frac{\sqrt{3}}{3}\int_0^1 t\,(e^{3t})'dt =$$

$$= \frac{\sqrt{3}}{3}\left(te^{3t}\Big|_0^1 - \int_0^1 e^{3t}dt\right) = \frac{\sqrt{3}}{3}\left(1\cdot e^3 - \int_0^1 \frac{1}{3}\cdot 3e^{3t}dt\right) =$$

$$= \frac{\sqrt{3}}{3}\left(e^3 - \frac{1}{3}\int_0^1 (e^{3t})'dt\right) = \frac{\sqrt{3}}{3}\left(e^3 - \frac{1}{3}\cdot e^{3t}\Big|_0^1\right) = \frac{\sqrt{3}}{3}\left(e^3 - \frac{1}{3}\cdot e^3 + \frac{1}{3}\right) =$$

$$= \frac{\sqrt{3}}{9}(2e^3+1).$$

② $\int_\gamma ye^{-x}ds$, unde parametrizarea curbei este:

$$\gamma(t) = \begin{cases} x(t) = \ln(1+t^2) \\ y(t) = 2\operatorname{arctg} t - t \end{cases} ; t\in[0,1].$$

$$\gamma'(t) = \begin{cases} x'(t) = \frac{2t}{1+t^2} \\ y'(t) = \frac{2}{1+t^2} - 1 = \frac{1-t^2}{1+t^2} \end{cases} ; t\in[0,1]$$

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$\|\gamma'(t)\| = \sqrt{\left(\frac{2t}{1+t^2}\right)^2 + \left(\frac{1-t^2}{1+t^2}\right)^2} = \frac{1}{1+t^2}\sqrt{4t^2+1-2t^2+t^4} =$

$= \frac{1}{1+t^2}\sqrt{(t^2+1)^2} = \frac{t^2+1}{t^2+1} = 1$

$\Rightarrow \int_\gamma y e^{-x} ds = \int_0^1 1\cdot(2\operatorname{arctg} t - t)\, e^{-\ln(1+t^2)} dt = \int_0^1 (2\operatorname{arctg} t - t)\frac{1}{1+t^2} dt$

$= 2\int_0^1 \frac{\operatorname{arctg} t}{1+t^2} dt - \int_0^1 \frac{t}{1+t^2} dt = \frac{\pi^2}{16} - \frac{1}{2}\ln 2.$

$\int_0^1 \frac{\operatorname{arctg} t}{1+t^2} dt = \int_0^{\pi/4} u\, du = \frac{u^2}{2}\Big|_0^{\pi/4} = \frac{\pi^2}{32}$

$u = \operatorname{arctg} t \Rightarrow du = \frac{1}{1+t^2} dt;\ t=0 \Rightarrow u=0;\ t=1 \Rightarrow u = \frac{\pi}{4}$

$\int_0^1 \frac{t}{1+t^2} dt = \int_1^2 \frac{1}{2}\cdot\frac{1}{v} dv = \frac{1}{2}\ln v\Big|_1^2 = \frac{1}{2}\ln 2$

$v = 1+t^2 \Rightarrow dv = 2t\,dt \Rightarrow \frac{1}{2}dv = t\,dt;\ t=0 \Rightarrow v=1;\ t=1 \Rightarrow v=2$

(3.) $\int_\gamma xyz\, ds$, unde parametrizarea curbei este

$$\gamma(t) = \begin{cases} x(t) = t \\ y(t) = \frac{2}{3}\sqrt{2t^3} \\ z(t) = \frac{1}{2}t^2 \end{cases};\ t \in [0,1]$$

$$\gamma'(t) = \begin{cases} x'(t) = 1 \\ y'(t) = \frac{2\sqrt{2}}{3}\cdot\frac{3}{2}t^{\frac{1}{2}} = \sqrt{2t} \\ z'(t) = t \end{cases};\ t \in [0,1]$$

$\|\gamma'(t)\| = \sqrt{1^2 + 2t + t^2} = \sqrt{(t+1)^2} = t+1$

$\Rightarrow \int_\gamma xyz\, ds = \int_0^1 (t+1)\, t\, \frac{2}{3}\sqrt{2t^3}\cdot\frac{1}{2}t^2 dt = \frac{\sqrt{2}}{3}\int_0^1 t^4\sqrt{t}(t+1) dt =$

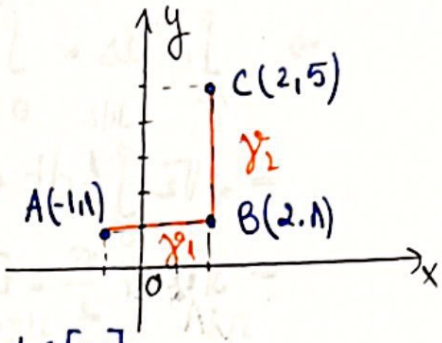
$= \frac{\sqrt{2}}{3}\int_0^1 t^5\sqrt{t} dt + \frac{\sqrt{2}}{3}\int_0^1 t^4\sqrt{t} dt = \frac{\sqrt{2}}{3}\left(\frac{2}{13} + \frac{2}{11}\right) = \frac{16\sqrt{2}}{143}.$

$\int_0^1 t^5\sqrt{t} dt = \int_0^1 t^{\frac{11}{2}} dt = \frac{2t^{\frac{13}{2}}}{13}\Big|_0^1 = \frac{2}{13}$

$\int_0^1 t^4\sqrt{t} dt = \int_0^1 t^{\frac{9}{2}} dt = \frac{2t^{\frac{11}{2}}}{11}\Big|_0^1 = \frac{2}{11}$

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④ $\int_\gamma x^2y\,ds$, unde curba γ este formată din segmentele AB și BC, cu $A(-1,1)$; $B(2,1)$; $C(2,5)$.



$$\int_\gamma x^2y\,ds = \int_{\gamma_1} x^2y\,ds + \int_{\gamma_2} x^2y\,ds.$$

- $\gamma_1(t) = \begin{cases} x(t) = (1-t)(-1) + t\cdot 2 = -1+t+t2 = 3t-1 \\ y(t) = (1-t)\cdot 1 + t\cdot 1 = 1-t+t = 1 \end{cases}; t\in[0,1]$

$\gamma_1'(t) = \begin{cases} x'(t) = 3 \\ y'(t) = 0 \end{cases} \Rightarrow \|\gamma_1'(t)\| = \sqrt{3}; t\in[0,1]$

$$\int_{\gamma_1} x^2y\,ds = \int_0^1 3(3t-1)^2\,dt = \sqrt{3}\int_0^1 (9t^2-6t+1)\,dt = \sqrt{3}(3t^3-3t^2+t)\Big|_0^1 =$$
$$= \sqrt{3}(3-3+1) = \sqrt{3}$$

- $\gamma_2(t) = \begin{cases} x(t) = (1-t)\cdot 2 + t\cdot 2 = 2-2t+2t = 2 \\ y(t) = (1-t)\cdot 1 + t\cdot 5 = 1-t+5t = 4t+1 \end{cases}; t\in[0,1]$

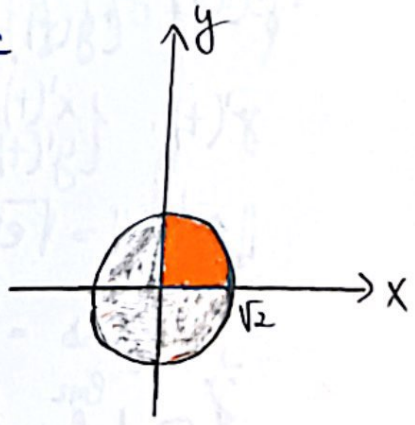
$\gamma_2'(t) = \begin{cases} x'(t) = 0 \\ y'(t) = 4 \end{cases} \Rightarrow \|\gamma_2'(t)\| = 4; t\in[0,1]$

$$\Rightarrow \int_{\gamma_2} x^2y\,ds = 2\int_0^1 8\cdot(4t+1)\,dt = 2\int_0^1 32t\,dt + 2\int_0^1 8\,dt = (32t^2+16t)\Big|_0^1 = 48$$

$$\Rightarrow \int_\gamma x^2y\,ds = 3+48 = 51.$$

⑤ $\int_\gamma x^2\,ds$, unde $\gamma: x^2+y^2=2$, $x\geq 0$; $y\geq 0$.

$x^2+y^2 = (\sqrt{2})^2$ este ecuația unui cerc centrat în origine cu raza $R=\sqrt{2}$.



$\gamma(t)$: $\begin{cases} x(t) = \sqrt{2}\cos t \\ y(t) = \sqrt{2}\sin t \end{cases}; t\in[0,\pi/2]$.

$\gamma'(t)$: $\begin{cases} x'(t) = -\sqrt{2}\sin t \\ y'(t) = \sqrt{2}\cos t \end{cases}; t\in[0,\pi/2]$.

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$\|\gamma'(t)\| = \sqrt{2\sin^2 t + 2\cos^2 t} = \sqrt{2}$

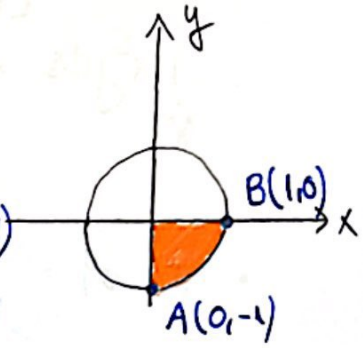
$\Rightarrow \int_\gamma x^2 ds = \int_0^{\pi/2} \sqrt{2}\cdot 2\cos^2 t\, dt = 2\sqrt{2}\int_0^{\pi/2} \cos^2 t\, dt = 2\sqrt{2}\int_0^{\pi/2} \frac{1+\cos 2t}{2}\, dt$

$= 2\sqrt{2}\int_0^{\pi/2} \frac{1}{2} dt + \frac{2\sqrt{2}}{2}\int_0^{\pi/2} \cos 2t\, dt = 2\sqrt{2}\cdot\frac{\pi}{2} + \sqrt{2}\ \frac{\sin 2x}{2}\Big|_0^{\pi/2} =$

$= \pi\sqrt{2} + \frac{\sqrt{2}}{2} - \frac{\sqrt{2}}{2}(2\pi+1)$.

(6) $\int_\gamma (x^2+y^2) ds$, unde γ: arcul de cerc $x^2+y^2=1$ de la $A(0,-1)$ la $B(1,0)$

$x^2+y^2=1$ este ecuația cercului centrat în origine, de rază $R=1$.



$\gamma(t): \begin{cases} x(t) = \cos t \\ y(t) = \sin t \end{cases}; t\in\left[\frac{3\pi}{2}, 2\pi\right]$ (cadranul IV)

$\gamma'(t): \begin{cases} x'(t) = -\sin t \\ y'(t) = \cos t \end{cases}; t\in\left[\frac{3\pi}{2}, 2\pi\right]$

$\|\gamma'(t)\| = \sqrt{\sin^2 t + \cos^2 t} = 1$

$\int_\gamma (x^2+y^2) ds = \int_{3\pi/2}^{2\pi} (\sin^2 t + \cos^2 t) dt = \int_{3\pi/2}^{2\pi} 1\, dt = 2\pi - \frac{3\pi}{2} = \frac{\pi}{2}$

(7) $\int_\gamma (x+y) ds$, unde parametrizarea arcului este

$\gamma(t): \begin{cases} x(t) = e^t + 1 \\ y(t) = e^t - 1 \end{cases}; t\in[0, \ln 2]$.

$\gamma'(t): \begin{cases} x'(t) = e^t \\ y'(t) = e^t \end{cases}; t\in[0, \ln 2]$

$\|\gamma'(t)\| = \sqrt{e^{2t}+e^{2t}} = e^t\sqrt{2}$

$\int_\gamma (x+y) ds = \int_0^{\ln 2} (e^t+1+e^t-1) e^t\sqrt{2}\, dt = \int_0^{\ln 2} 2e^{2t}\sqrt{2}\, dt = 2\sqrt{2}\int_0^{\ln 2} e^{2t} dt =$

$= 2\sqrt{2}\cdot\frac{1}{2}\int_0^{\ln 2} 2e^{2t} dt = \sqrt{2}\int_0^{\ln 2} (e^{2t})' dt = \sqrt{2}\, e^{2t}\Big|_0^{\ln 2} = \sqrt{2}(4-1) = 3\sqrt{2}$.

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(8) $\int_\gamma (2x-y)\,ds$, unde parametrizarea curbei este:

$$\gamma(t): \begin{cases} x(t) = \sin t \\ y(t) = -\cos t \end{cases}; t \in \left[0, \frac{\pi}{2}\right]$$

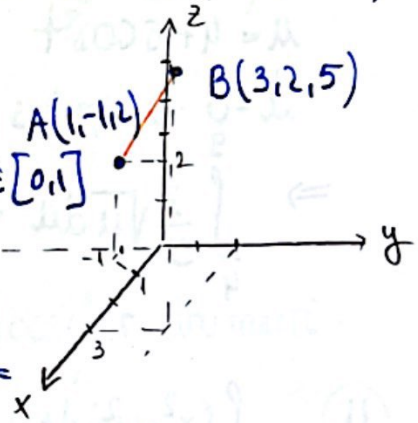
$$\gamma'(t): \begin{cases} x'(t) = \cos t \\ y'(t) = -\sin t \end{cases}; t \in \left[0, \frac{\pi}{2}\right]$$

$$\|\gamma'(t)\| = \sqrt{\cos^2 t + \sin^2 t} = 1$$

$$\Rightarrow \int_\gamma (2x-y)\,ds = \int_0^{\pi/2} (2\sin t - \cos t)\,dt = 2\int_0^{\pi/2} \sin t\,dt - \int_0^{\pi/2} \cos t\,dt =$$

$$= 2\cdot(-\cos t)\Big|_0^{\pi/2} - \sin t\Big|_0^{\pi/2} = 2(0+1) - (1-0) = 1.$$

(9) $\int_\gamma xyz\,ds$, unde γ: segmentul AB, cu $A(1,-1,2)$ și $B(3,2,5)$



$$\gamma(t): \begin{cases} x(t) = (1-t)\cdot 1 + t\cdot 3 = 1-t+3t = 2t+1 \\ y(t) = (1-t)\cdot(-1) + t\cdot 2 = -1+t+2t = 3t-1 \\ z(t) = (1-t)\cdot 2 + t\cdot 5 = 2-2t+5t = 3t+2 \end{cases}; t \in [0,1]$$

$$\gamma'(t): \begin{cases} x'(t) = 2 \\ y'(t) = 3 \\ z'(t) = 3 \end{cases}; t \in [0,1] \Rightarrow \|\gamma'(t)\| = \sqrt{4+9+9} = \sqrt{22}$$

$$\int_\gamma xyz\,ds = \int_0^1 \sqrt{22}(2t+1)(3t-1)(3t+2)\,dt = \sqrt{22}\int_0^1 (6t^2+t-1)(3t+2)\,dt =$$

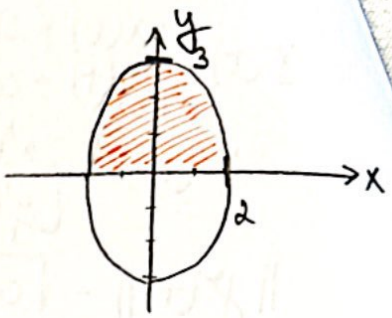
$$= \int_0^1 \sqrt{22}\left(18t^3+3t^2-3t+12t^2+2t-2\right)dt = \sqrt{22}\int_0^1 (18t^3+15t^2-t-2)\,dt =$$

$$= \sqrt{22}\left(\frac{18}{4}t^4 + 5t^3 - \frac{1}{2}t^2 - 2t\right)\Big|_0^1 = \sqrt{22}\left(\frac{9}{2}\cdot 1 + 5 - \frac{1}{2} - 2\right) =$$

$$= 7\sqrt{22}.$$

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(10.) $\int_\gamma xy\,ds$, unde $\gamma: \frac{x^2}{4}+\frac{y^2}{9}=1; y\geq 0$



$\frac{x^2}{2^2}+\frac{y^2}{3^2}=1$ este ec unei elipse de semiaxe $a=2$ și $b=3$, centrată în origine.

$\gamma(t): \begin{cases} x(t)=2\cos t \\ y(t)=3\sin t \end{cases}; t\in[0,\pi]$

$\gamma'(t): \begin{cases} x'(t)=-2\sin t \\ y'(t)=3\cos t \end{cases}; t\in[0,\pi]$

$\Rightarrow \|\gamma'(t)\| = \sqrt{4\sin^2 t+9\cos^2 t} = \sqrt{4+5\cos^2 t}$

$\int_\gamma xy\,ds = \int_0^\pi 6\sin t\cos t\sqrt{4+5\cos^2 t}\,dt$

$u=4+5\cos^2 t \Rightarrow du=-5\cdot 2\cos t\sin t\,dt \Rightarrow \sin t\cos t\,dt=\frac{-1}{10}du$

$t=0\Rightarrow u=9$; $t=\pi\Rightarrow u=4$

$\Rightarrow \int_4^9 \frac{3}{5}\sqrt{u}\,du = \frac{3}{5}\int_4^9 u^{\frac{1}{2}}du = \frac{3}{5}\cdot\frac{2u\sqrt{u}}{3}\Big|_4^9 = \frac{2}{5}(27-8)=\frac{38}{5}$

(11.) $\int_\gamma (x^2+y^2)\,ds$, unde parametrizarea curbei este

$\gamma(t): \begin{cases} x(t)=4t-1 \\ y(t)=3t+1 \end{cases}; t\in[-1,1]$

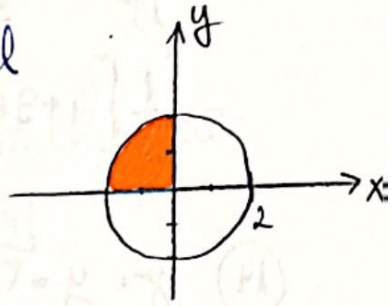
$\gamma'(t): \begin{cases} x'(t)=4 \\ y'(t)=3 \end{cases}; t\in[-1,1] \Rightarrow \|\gamma'(t)\| = \sqrt{4^2+3^2}=5.$

$\int_\gamma (x^2+y^2)\,ds = \int_{-1}^1 5(4t-1)(3t+1)\,dt = 5\int_{-1}^1 (12t^2+t-1)\,dt =$

$= 5\left(4t^3+\frac{1}{2}t^2-t\right)\Big|_{-1}^1 = 5\left(4+\frac{1}{2}-1+4-\frac{1}{2}-1\right)=30.$

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(12.) $\int_\gamma y^2 ds$, unde $\gamma: x^2+y^2=4;\ x\le 0;\ y\ge 0$



$x^2+y^2=4$ este ecuația cercului cu centrul în origine și cu raza $R=2$.

$$\gamma(t): \begin{cases} x(t) = 2\cos t \\ y(t) = 2\sin t \end{cases}; t\in\left[\frac{\pi}{2},\pi\right]$$

$$\gamma'(t): \begin{cases} x'(t) = -2\sin t \\ y'(t) = 2\cos t \end{cases}; t\in\left[\frac{\pi}{2},\pi\right]$$

$$\Rightarrow \|\gamma'(t)\| = \sqrt{4\sin^2 t+4\cos^2 t} = \sqrt{4} = 2$$

$$\Rightarrow \int_\gamma y^2 ds = \int_{\pi/2}^{\pi} 4\sin^2 t\cdot 2dt = 8\int_{\pi/2}^{\pi}\sin^2 t\,dt = 8\int_{\pi/2}^{\pi}\frac{1-\cos 2t}{2}dt =$$

$$= 8\int_{\pi/2}^{\pi}\frac{1}{2}dt - 8\cdot\frac{1}{2}\int_{\pi/2}^{\pi}\cos 2t\,dt = 4\left(\pi-\frac{\pi}{2}\right) - 4\int_{\pi/2}^{\pi}\left(\frac{\sin 2t}{2}\right)'dt =$$

$$= \frac{4\pi}{2} - 4\frac{\sin 2t}{2}\Big|_{\pi/2}^{\pi} = 2\pi - 2(0-0) = 2\pi.$$

Să se calculeze lungimea următoarelor drumuri:

(13.) $$\gamma(t): \begin{cases} x(t) = e^t \\ y(t) = e^{3t/2} \end{cases}; t\in[0,1]$$

$$\gamma'(t): \begin{cases} x'(t) = e^t \\ y'(t) = \frac{3}{2}e^{\frac{3t}{2}} \end{cases}; t\in[0,1]$$

$$\|\gamma'(t)\| = \sqrt{e^{2t}+\frac{9}{4}e^{3t}} = \frac{1}{2}e^t\sqrt{4+9e^t}$$

$$\Rightarrow \int_\gamma ds = \int_0^1 \frac{1}{2}e^t\sqrt{4+9e^t}dt = \frac{1}{2}\int_0^1 (e^t)'\cdot\sqrt{4+9e^t}dt =$$

$$= \frac{1}{2}\left(e^t\sqrt{4+9e^t}\Big|_0^1 - \int_0^1 e^t\cdot\frac{9e^t}{2\sqrt{4+9e^t}}dt\right) =$$

$$\frac{1}{2}\int_0^1 \frac{9e^t\cdot e^t}{\sqrt{4+9e^t}}dt = \frac{1}{2}\int_{13}^{4+9e}\frac{u-4}{9\sqrt{u}}du = \frac{1}{18}\int_{13}^{4+9e}\sqrt{u}\,du - \frac{2}{9}\int_{13}^{4+9e}\frac{1}{\sqrt{u}}du =$$

$4+9e^t = u \Rightarrow 9e^t dt = du \Rightarrow e^t = \frac{u-4}{9};\ t=0\Rightarrow u=13;\ t=1\Rightarrow u=4+9e$

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$$= \frac{1}{9}\cdot\frac{2u\sqrt{u}}{3}\Big|_{13}^{4+9\ell} - \frac{2}{9}\cdot 2\sqrt{u}\Big|_{13}^{4+9\ell} =$$

$$= \frac{1}{27}\left[(4+9\ell)^{3/2} - 13^{3/2}\right] - \frac{4}{9}\left[(4+9\ell)^{1/2} - 13^{1/2}\right] \dots$$

(14) $\gamma: y = \ln\sin x \,;\, x\in\left[\frac{\pi}{6}, \frac{\pi}{3}\right]$

$$\gamma(t): \begin{cases} x(t) = t \\ y(t) = \ln\sin(t) \end{cases}; t\in\left[\frac{\pi}{6}, \frac{\pi}{3}\right]$$

$$\gamma'(t): \begin{cases} x'(t) = 1 \\ y'(t) = \frac{1}{\sin t}\cdot\cos t = \operatorname{ctg} t \end{cases}; t\in\left[\frac{\pi}{6}, \frac{\pi}{3}\right]$$

$$\|\gamma'(t)\| = \sqrt{\frac{\cos^2 t}{\sin^2 t} + 1} = \sqrt{\frac{1}{\sin^2 t}} = \frac{1}{\sin t} \quad \pi/3 \;|\; \pi/6$$

$$L = \int_\gamma ds = \int_{\pi/6}^{\pi/3} \frac{1}{\sin t}\,dt = \int_{\pi/6}^{\pi/3} \frac{\sin t}{1-\cos^2 t}\,dt$$

$\cos t = u \Rightarrow -\sin t\,dt = du \Rightarrow \sin t\,dt = -du$

$t = \frac{\pi}{6} \Rightarrow u = \frac{\sqrt{3}}{2}$; $t = \frac{\pi}{3} \Rightarrow u = \frac{1}{2}$ $\quad \sqrt{3}/2 \;|\; 3$

$$\Rightarrow L = \int_{1/2}^{\sqrt{3}/2} \frac{1}{1-u^2}\,du = -\int_{1/2}^{\sqrt{3}/2} \frac{1}{u^2-1}\,du = -\frac{1}{2}\ln\frac{u-1}{u+1}\Big|_{1/2}^{\sqrt{3}/2} =$$

$$= -\frac{1}{2}\left[\ln\frac{\frac{\sqrt{3}}{2}-1}{\frac{\sqrt{3}}{2}+1} - \ln\frac{\frac{1}{2}-1}{\frac{1}{2}+1}\right] = -\frac{1}{2}\left[\ln\frac{\sqrt{3}-2}{\sqrt{3}+2} - \ln\frac{-\frac{1}{2}}{\frac{3}{2}}\right] =$$

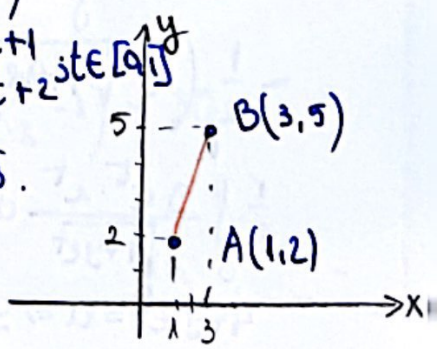
$$= \frac{1}{2}\left[\ln\frac{\sqrt{3}-2}{\sqrt{3}+2} + \ln 3\right].$$

(15) γ: segmentul AB, cu A(1,2) și B(3,5)

$$\gamma(t): \begin{cases} x(t) = (1-t)\cdot 1 + t\cdot 3 = 1 - t + 3t = 2t+1 \\ y(t) = (1-t)\cdot 2 + t\cdot 5 = 2 - 2t + 5t = 3t+2 \end{cases}; t\in[0,1]$$

$$\gamma'(t): \begin{cases} x'(t) = 2 \\ y'(t) = 3 \end{cases}; t\in[0,1]; \|\gamma'(t)\| = \sqrt{13}.$$

$$L = \int_\gamma ds = \int_0^1 \sqrt{13} = \sqrt{13}$$



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Să se calculeze $\int_\gamma \alpha$ (integrale curbilinii de speța a doua):

(16) $\alpha = (x^2+y^2)dx + (x^2-y^2)dy$, unde parametrizarea curbei este $\gamma(t): \begin{cases} x(t) = \sqrt{t} \\ y(t) = \sqrt{t+1} \end{cases}; t \in [1,4]$

$\gamma'(t): \begin{cases} x'(t) = \frac{1}{2\sqrt{t}} \\ y'(t) = \frac{1}{2\sqrt{t+1}} \end{cases}; t \in [1,4]$

$P(x,y) = x^2+y^2$; $Q(x,y) = x^2-y^2$

$$I = \int_\gamma \alpha = \int_1^4 \left[(t+t+1)\cdot\frac{1}{2\sqrt{t}} + (t-t-1)\frac{1}{2\sqrt{t+1}} \right] dt =$$

$$= \frac{1}{2}\int_1^4 \left(\frac{2t+1}{\sqrt{t}} - \frac{1}{\sqrt{t+1}} \right) dt = \frac{1}{2}\int_1^4 \frac{2t+1}{\sqrt{t}}dt - \frac{1}{2}\int_1^4 \frac{1}{\sqrt{t+1}}dt =$$

$$= \frac{17}{3} - \sqrt{5} + \sqrt{2}.$$

$$\int_1^4 \frac{2t+1}{\sqrt{t}}dt = 2\int_1^4 \sqrt{t}dt + \int_1^4 \frac{1}{\sqrt{t}}dt = 2\,\frac{2t\sqrt{t}}{3}\Big|_1^4 + 2\sqrt{t}\Big|_1^4 =$$

$$= \frac{4}{3}(8-1) + 2(2-1) = \frac{28}{3} + 2 = \frac{34}{3}$$

$$\int_1^4 \frac{1}{\sqrt{t+1}}dt = \int_2^5 \frac{1}{\sqrt{u}}du = 2\sqrt{u}\Big|_2^5 = 2(\sqrt{5}-\sqrt{2})$$

$t+1 = u \Rightarrow dt = du$; $t=1 \Rightarrow u=2$; $t=4 \Rightarrow u=5$

(17) $\alpha = \frac{1}{1+y^2}dx + \frac{y}{1+x^2}dy$, unde parametrizarea este $\gamma(t): \begin{cases} x(t) = t^2 \\ y(t) = t \end{cases}; t \in [0,1]$

$\gamma'(t): \begin{cases} x'(t) = 2t \\ y'(t) = 1 \end{cases}; t \in [0,1]$

$P(x,y) = \frac{1}{1+y^2}$; $Q(x,y) = \frac{y}{1+x^2}$

$$I = \int_\gamma \alpha = \int_0^1 \left(\frac{1}{1+t^2}\cdot 2t + \frac{t}{1+t^4} \right) dt = \int_0^1 \frac{2t}{1+t^2}dt + \int_0^1 \frac{t}{1+t^4}dt$$

$$= \ln 2 + \frac{\pi}{8}.$$

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$$\int_0^1 \frac{2t}{1+t^2}dt = \int_1^2 \frac{1}{u}du = \ln u\Big|_1^2 = \ln 2$$

$1+t^2 = u \Rightarrow 2tdt = du$; $t=0 \Rightarrow u=1$; $t=1 \Rightarrow u=2$

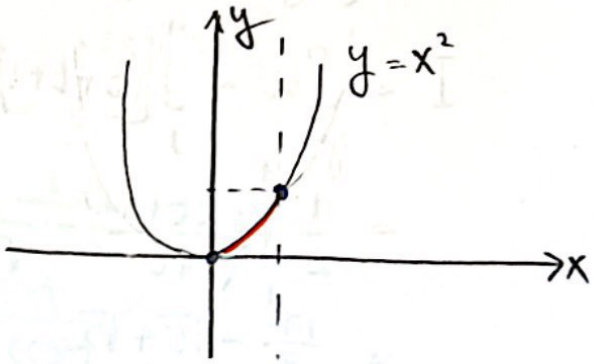
$$\int_0^1 \frac{t}{1+t^4}dt = \frac{1}{2}\int_0^1 \frac{1}{1+v^2}dv = \frac{1}{2}\operatorname{arctg} v\Big|_0^1 = \frac{1}{2}\cdot\frac{\pi}{4} = \frac{\pi}{8}$$

$t^2 = v \Rightarrow 2tdt = dv \Rightarrow tdt = \frac{1}{2}dv$; $t=0 \Rightarrow v=0$; $t=1 \Rightarrow v=1$

18. $\alpha = \sqrt{x}dx + xy^2dy$, unde $\gamma: y = x^2$; $x \in [0,1]$

pt $y = x^2$, $x_{vf} = 0$; $y_{vf} = 0$

$x = 1 \Rightarrow y = 1$



$$\gamma(t): \begin{cases} x(t) = t \\ y(t) = t^2 \end{cases}; t \in [0,1]$$

$$\gamma'(t): \begin{cases} x'(t) = 1 \\ y'(t) = 2t \end{cases}; t \in [0,1]$$

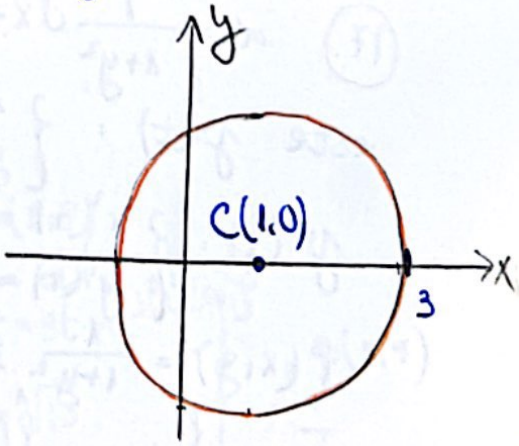
$P(x,y) = \sqrt{x}$; $Q(x,y) = xy^2$

$$I = \int_\gamma \alpha = \int_0^1 \left(\sqrt{t} + t^5 \cdot 2t\right)dt = \int_0^1 \sqrt{t}dt + 2\int_0^1 t^6 dt =$$

$$= \frac{2t\sqrt{t}}{3}\Big|_0^1 + 2\,\frac{t^7}{7}\Big|_0^1 = \frac{2}{3} + \frac{2}{7} = \frac{20}{21}.$$

19. $\alpha = (1-y)dx + (x-1)dy$, unde $\gamma: x^2+y^2-2x = 3$

$x^2+y^2-2x = 3 \Leftrightarrow (x-1)^2 + y^2 = (2)^2$ este ecuatia cercului centrat in $(1,0)$ cu raza $R = 2$.



$$\gamma(t): \begin{cases} x(t) = 1 + 2\cos t \\ y(t) = 2\sin t \end{cases}; t \in [0, 2\pi]$$

$$\gamma'(t): \begin{cases} x'(t) = -2\sin t \\ y'(t) = 2\cos t \end{cases}; t \in [0, 2\pi]$$

$P(x,y) = 1-y$; $Q(x,y) = x-1$.

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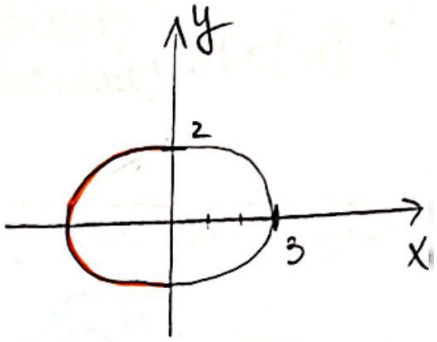
$$I = \int_{\gamma_{2\pi}} \alpha = \int_0^{2\pi} \left[(1-2\sin t)(-2\sin t) + (1+2\cos t-1)\cdot 2\cos t\right] dt =$$

$$= \int_0^{2\pi} \left(4\sin^2 t - 2\sin t + 4\cos^2 t\right) dt = \int_0^{2\pi} (4 - 2\sin t)\, dt =$$

$$= 2\int_0^{2\pi} 2\,dt - 2\int_0^{2\pi} \sin t\, dt = 4\cdot 2\pi - 2(-\cos t)\Big|_0^{2\pi} = 8\pi$$

(20.) $\alpha = (4x-6y)dx + (6x+9y)dy$, unde $\gamma: \frac{x^2}{9} + \frac{y^2}{4} = 1; x \le 0$

$\frac{x^2}{9} + \frac{y^2}{4} = 1$ este ecuația unei elipse de semiaxe $a=3$ și $b=2$ centrată în origine.



$$\gamma(t): \begin{cases} x(t) = 3\cos t \\ y(t) = 2\sin t \end{cases}; t \in \left[\frac{\pi}{2}, \frac{3\pi}{2}\right]$$

$$\gamma'(t): \begin{cases} x'(t) = -3\sin t \\ y'(t) = 2\cos t \end{cases}; t \in \left[\frac{\pi}{2}, \frac{3\pi}{2}\right]; \quad P(x,y) = 4x-6y, \quad Q(x,y) = 6x+9y$$

$$I = \int_\gamma \alpha = \int_{\pi/2}^{3\pi/2} \left[(12\cos t - 12\sin t)(-3\sin t) + (18\cos t + 18\sin t)2\cos t\right] dt$$

$$= \int_{\pi/2}^{3\pi/2} \left[-36(\cos t - \sin t)\sin t + 36(\cos t + \sin t)\cos t\right] dt =$$

$$= \int_{\pi/2}^{3\pi/2} \left(-\cancel{36\sin t\cos t} + 36\sin^2 t + 36\cos^2 t + \cancel{36\sin t\cos t}\right) dt =$$

$$= \int_{\pi/2}^{3\pi/2} 36\, dt = 36\left(\frac{3\pi}{2} - \frac{\pi}{2}\right) = 36\pi.$$

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